

TECHNICAL CASE STUDY

Mineral Prospectivity Mapping from Satellite Imagery

Binary segmentation of mineral-bearing zones from Landsat-8 multispectral data over Salem District, Tamil Nadu — comparing three deep learning architectures under extreme class imbalance.

- github.com/anitkumarmaity1-hash/Minerals_Prospective_Mapping

95.11%

BEST TEST F1-SCORE

92.4%

BEST TEST RECALL

2,509:1

BACKGROUND-TO-MINERAL RATIO

3

ARCHITECTURES COMPARED

Anit Kumar Maity

MCA, Pondicherry University · Dept. of Computer Science

U-NET · ATTENTION U-NET · RESUNET

01 — THE PROBLEM

Mineral zones are geologically rare and spectrally noisy

- 01 Spectral overlap.** Band ratios and PCA alone can't separate mineral signatures from background rock and soil.
- 02 No spatial context.** Pixel-wise statistical methods ignore neighbouring-pixel relationships that CNNs capture naturally.
- 03 Severe imbalance.** Verified mineral occurrences cover a tiny fraction of the study area — most segmentation losses collapse to predicting "background" everywhere.
- 04 Field surveys are expensive.** A reliable prospectivity map narrows the search area before geologists ever visit the site.

SALEM DISTRICT · VALID PIXELS

5,794,980 valid pixels

Landsat-8, 30m, cropped to boundary



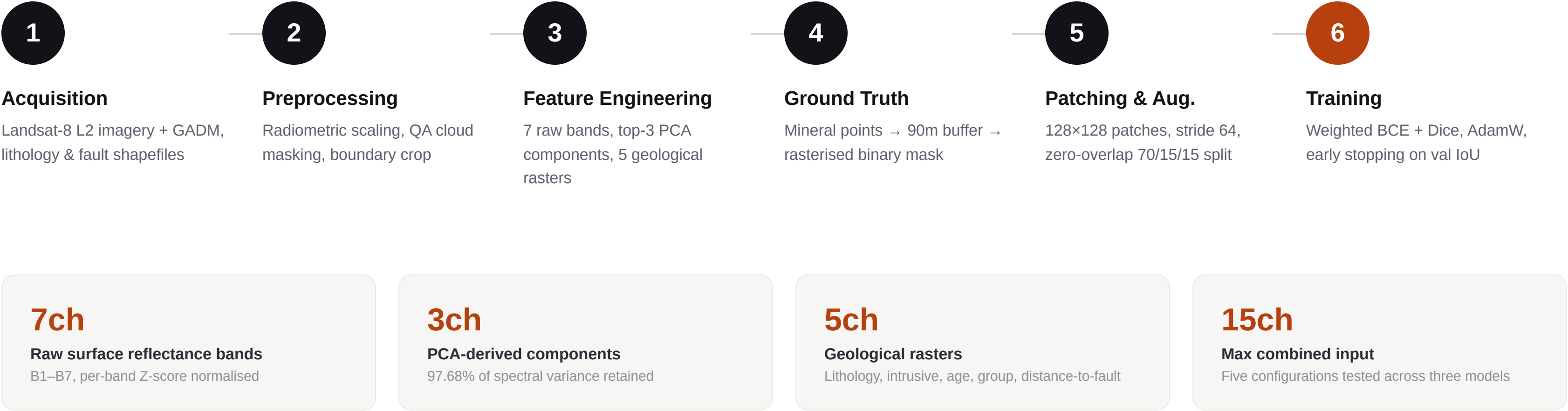
2,309 labelled mineral pixels

0.04%

2,509:1

background-to-mineral pixel ratio — drives every loss-weighting and augmentation decision downstream

One pipeline, five feature configurations, three models



Best configuration per model, held-out test set

MODEL	BEST INPUT / MECHANISM	TEST IOU	TEST F1	TEST RECALL
U-Net	3-channel PCA input, deep supervision on two auxiliary decoder heads	90.67%	95.11%	92.33%
Attention U-Net	8-channel PCA + geological rasters, soft attention gates on every skip connection	88.84%	94.09%	90.25%
ResUNet ★	15-channel full feature stack, residual blocks with identity shortcuts	88.51%	93.90%	92.42%

Loss: Weighted BCE (pos_weight = 45) + Dice, equal-weighted · **Optimiser:** AdamW, ReduceLROnPlateau · **Stopping:** early stopping on validation Mineral IoU, patience 15

A missed deposit costs more than a false alarm

Under a **2,509:1** background-to-mineral imbalance, IoU and F1 are precision-dominated — they penalise a false positive and a false negative equally. In mineral exploration, that trade-off is backwards.

A false positive costs a short field visit. A false negative means a real deposit is **never investigated** — a permanent, irreversible loss.

ResUNet was selected as the operational model: highest recall (**92.42%**) on the richest 15-channel input, with residual connections preventing gradient degradation across the deeper, heterogeneous feature stack.

U-NET · HIGHEST IOU

90.67%

Best on paper — but 8 more missed mineral pixels than ResUNet on the test set

RESUNET · CHOSEN MODEL

92.42%

Highest recall — maximises mineral detection completeness for field survey planning

Full report, notebooks & code

github.com/anitkumarmaity1-hash/Minerals_Prospective_Mapping